

Adaptive Traffic Intelligence for High-Performance LPX Systems

Abstract

This project explores adaptive traffic management in a multi-LPU LPX architecture using VisualSim. Three models were progressively developed: the LPX Static Baseline Model, LPX Adaptive Stress-Aware Model, and LPX Adaptive Model.

The aim was to move from fixed resource mapping toward a system that can monitor LPU activity, identify workload variations, and support adaptive traffic decisions through feedback.

Problem Statement

In multi-resource processing systems, static traffic mapping can lead to uneven resource utilization, where certain LPUs experience higher activity while others remain comparatively underutilized.

The project addresses this limitation by introducing resource monitoring and adaptive traffic management, allowing the architecture to respond to changing workload conditions.

Model 1 — LPX Static Baseline

The Static Baseline Model establishes the reference architecture.

Traffic Input → Mapper → LPUs → Statistics

Incoming transactions are distributed among multiple LPUs using a predefined mapping strategy. The routing decision does not change according to the current condition of the processing resources.

The simulation showed that different LPUs could experience different levels of activity. Since the mapping remains fixed, the architecture cannot actively respond to these workload variations.

This model therefore establishes the main limitation addressed by the project: fixed routing lacks awareness of changing resource conditions.

Model 2 — LPX Adaptive Stress-Aware

The second model introduces resource monitoring and stress awareness.

Traffic Input → Mapper → LPUs → Statistics → Load Observation

Statistics from the individual LPUs are monitored to understand how the processing resources behave during simulation.

Different activity and mean values were observed across the LPUs, indicating that the workload was not identical across every processing resource.

Instead of treating these statistics only as outputs, the stress-aware architecture uses them as an indication of the relative condition of each LPU.

A highly active LPU can be identified as comparatively stressed, while another resource with lower activity may have more available processing capacity.

This introduces the key concept of understanding the state of the system before making traffic-management decisions.

Model 3 — LPX Adaptive

The final LPX Adaptive Model extends the architecture toward dynamic traffic management.

Traffic → Monitor → Decision Logic → Mapper → LPUs → Statistics → Feedback

The model combines resource monitoring with adaptive decision logic. Information collected from the processing resources can influence subsequent traffic-management decisions.

The adaptive process follows:

Monitor → Detect → Decide → Route → Measure → Feedback

This creates a closed-loop architecture, where the outcome of previous processing activity can influence future routing behavior.

The system therefore progresses from simply processing traffic to observing and responding to changing resource conditions.

VisualSim Implementation

The three models were implemented using VisualSim system-level modeling blocks, including:

- Traffic generation
- Mapper blocks
- Multiple LPU resources
- Statistics and monitoring
- Decision logic
- Adaptive control and feedback

The modular implementation allowed the architecture to evolve progressively without redesigning the entire system.

Results and Observations

The Static Baseline Model demonstrated the behavior of fixed resource mapping and showed that multiple LPUs can experience different levels of activity.

The Adaptive Stress-Aware Model introduced monitoring of these differences, allowing the architecture to identify comparatively busy and less-active resources.

The Adaptive Model connected monitoring and decision-making through feedback, demonstrating how resource information can support dynamic traffic-management decisions.

The overall progression achieved was:

Static Routing → Resource Monitoring → Stress Awareness → Adaptive Routing → Feedback

Key Innovation

The key innovation is the transition from fixed traffic mapping to closed-loop adaptive traffic management.

Instead of treating LPUs only as processing destinations, their behavior becomes part of the decision-making process.

This allows the architecture to potentially respond to changing workload conditions rather than continuing with the same predefined routing strategy.

Future Scope

The adaptive architecture can be extended with:

- Congestion prediction
- Queue and latency-aware routing
- Thermal and power-aware scheduling
- Machine-learning-based traffic decisions
- Fault-aware resource selection
- Scaling to larger LPX architectures

These improvements could enable the system to predict resource stress before congestion occurs and make more intelligent routing decisions.

Conclusion

This project demonstrates the evolution of an LPX architecture from static traffic mapping to adaptive, feedback-based resource management.

The first model established the static baseline, the second introduced LPU monitoring and stress awareness, and the third extended these concepts toward adaptive traffic control using feedback.

The different behavior observed across the LPUs highlighted the need for resource-aware traffic management.

The project ultimately demonstrates that monitoring processing resources and incorporating their state into routing decisions can provide a foundation for more intelligent, scalable, and congestion-aware LPX systems.